

GEOGRAPHY

PHYSICAL GEOGRAPHY



CLIMATOLOGY

MODULE – 46

WORLD CLIMATIC TYPES
THE CHINA TYPE CLIMATE

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GLOBAL CLIMATE CLASSIFICATION

CLIMATE ZONE	LATITUDE	CLIMATIC TYPE
Equatorial zone	0-10° N & S	1. Hot, Wet Equatorial climate
Hot Zone	10°-30° N & S	2. a) Tropical Monsoon b) Tropical Marine
		3. Sudan type
		4. Desert type a) Saharan type b) Mid-Latitude type
Warm Temperature Zone	30°-40° N & S	5. Western Margin Mediterranean Type
		6. Central Continental Steppe type.
		7. Eastern Margin a) China type b) Gulf type c) Natal type

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GLOBAL CLIMATE CLASSIFICATION

CLIMATE ZONE	LATITUDE	CLIMATIC TYPE
Cool Temperature Zone	45°-65° N & S	8. Western Margin - British type.
		9. Siberian type - Central Continental
		10. Laurentian type - Eastern Margin
Cold Zone	65°-90° N & S	11. Arctic or Polar or Tundra climate.

WORLD CLIMATIC TYPES

10

THE WARM TEMPERATE EASTERN MARGIN (CHINA TYPE) CLIMATE

DISTRIBUTION

1 This type of climate is found on the **eastern margins of continents in warm temperate latitudes**, just outside the tropics.

2 It has comparatively **more rainfall than the Mediterranean climate** in the same latitudes, coming mainly in the summer.

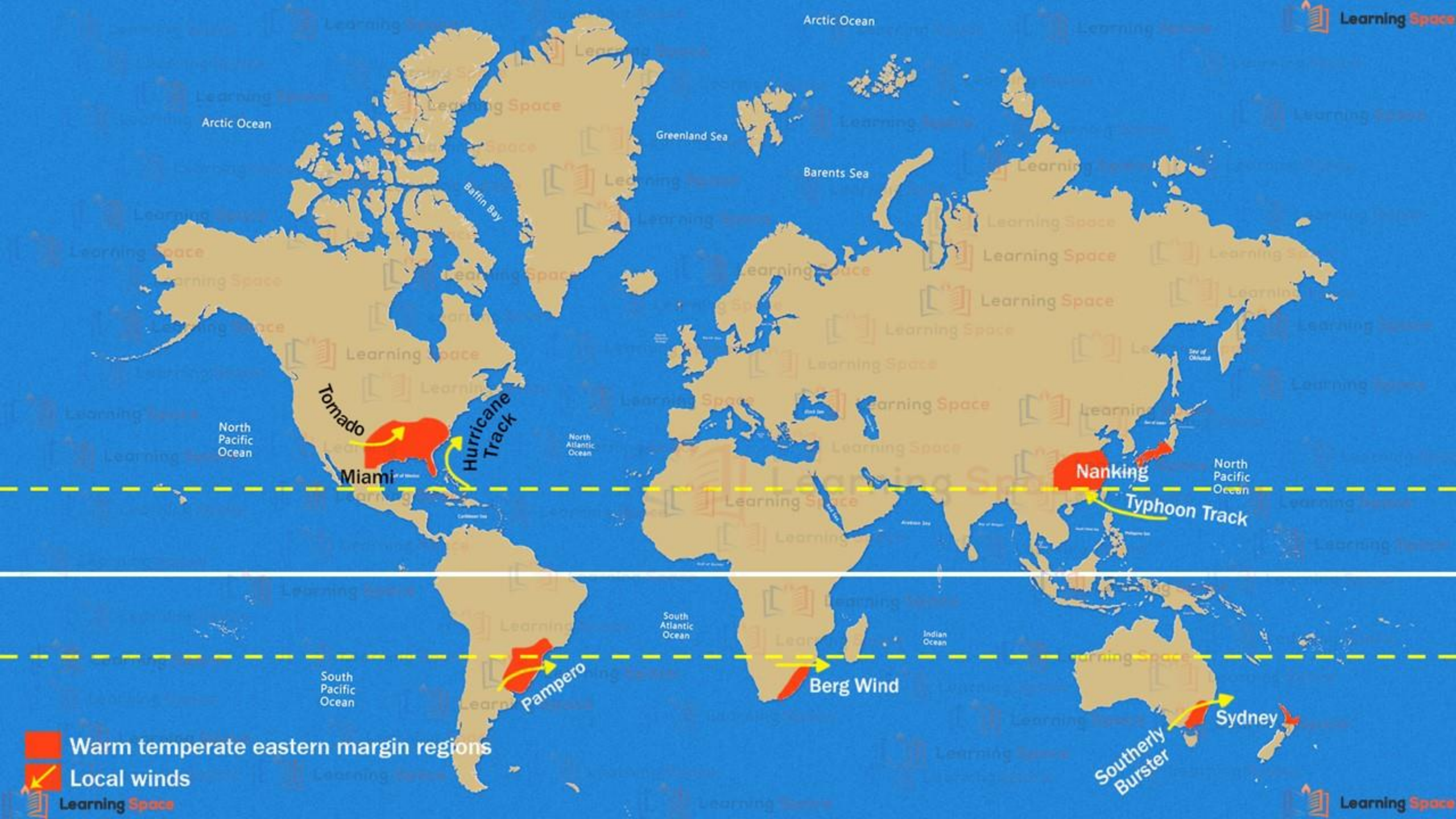
3 It is, in fact, the climate of most **parts of China- a modified form of monsoonal climate**.

4 It is thus also called the **Temperature Monsoon or China Type of climate**.

5 In south-eastern U.S.A., bordering the Gulf of Mexico, continental heating in summer induces an inflow of air from the cooler Atlantic Ocean.

6 Though less pronounced, the overall climatic features resemble those of the **China type**.

7 It is sometimes referred to as the **Gulf type of climate**.



Warm temperate eastern margin regions

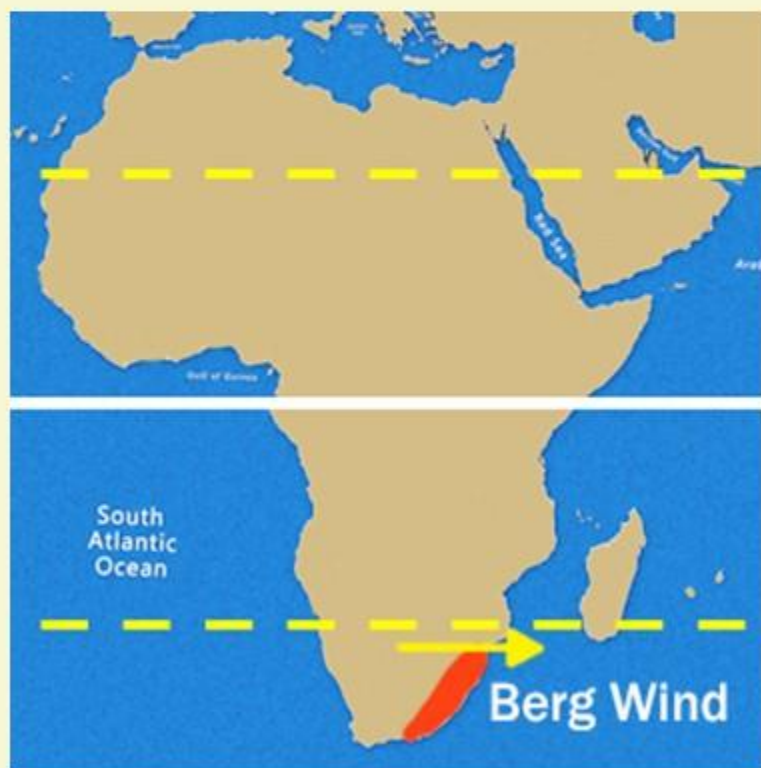
Local winds

WORLD CLIMATIC TYPES

10

THE WARM TEMPERATE EASTERN MARGIN (CHINA TYPE) CLIMATE

DISTRIBUTION



8

In the southern hemisphere, this kind of climate is experienced along the warm temperate eastern coastlands of all the three continents: in New South Wales with its eucalyptus forests; in Natal where cane sugar thrives; and in the maize belt of the Parana-Paraguay-Uruguay basin.

9

As the regions are influenced by the on-shore Trade Winds all the year round, without any monsoon variations, the climate cannot be described as temperate monsoon. It is sometimes referred to as the Natal type of climate.

10 THE CHINA TYPE CLIMATE

CLIMATE

- 1 The Warm Temperate Eastern Margin Climate is typified by a warm moist summer and a cool, dry winter.
- 2 The mean monthly temperature varies between 5°C and 25°C and is strongly modified by maritime influence.
- 3 Occasionally, the penetration of cold air from the continental interiors may bring down the temperature to freezing point.
- 4 Rainfall is more than moderate, anything from 60-150 cm.
- 5 This is an important feature i.e. the region receives fairly uniform distribution of rainfall throughout the year.
- 6 There is rain every month, except in the interior of central China, where there is a distinct dry season.
- 7 Rain comes either from convectional sources or as orographic rain in summer, or from depressions in prolonged showers in winter.

10 THE CHINA TYPE CLIMATE

CLIMATE



8 Local storms, e.g. typhoons, and hurricanes, also occur.

- 9 It can be sub-divided into three main types.
- **The China type**: central and north China, including southern Japan (temperate monsoonal).
 - **The Gulf type**: south-eastern United States, (slight-monsoonal).
 - **The Natal type**: all the warm temperate eastern margin areas of the southern hemisphere including Natal, eastern Australia and southern Brazil-Paraguay-Uruguay and northern Argentina (non-monsoonal).

10 THE CHINA TYPE CLIMATE

NATURAL VEGETATION



- 1 The eastern margins of warm temperate latitudes have **a much heavier rainfall than either the western margins or the continental interiors and thus have a luxuriant vegetation.**
- 2 The lowlands carry both ever-green broad-leaved forests and deciduous trees quite similar to those of the tropical monsoon forests.
- 3 On the highlands, are various species of conifers such as pines and cypresses which are important soft-woods.

10 THE CHINA TYPE CLIMATE

NATURAL VEGETATION

- 4 As the perennial plant growth is not checked by either a dry season as in the Mediterranean, or a cold season as in the cool temperate regions, **conditions are well suited to a rich variety of plant life including grass, ferns, lianas, bamboos, palms and forests.**
- 5 **The well distributed rainfall all the year round makes the region look green at all times.**
- 6 The forests of China and southern Japan also have considerable economic value and include oak, camphor, camelia and magnolia.
- 7 Unfortunately the **tremendous population pressure in the two countries** has caused much of the original forest to be cleared for fuel or crop cultivation.
- 8 **Deforestation** has resulted in many barren hill-slopes that are still feeling the impact of soil erosion.
- 9 Much of the forest cover has given way to the **cultivation of sub-tropical crops like cotton, maize and fruits.**

10 THE CHINA TYPE CLIMATE

ECONOMIC DEVELOPMENT



Coffee plantations in Brazil

- 1 The warm temperate eastern margins are the most productive parts of the middle latitudes.
- 2 There is adequate rainfall, no prolonged drought, and the cold season is warm enough for most crops to survive.
- 3 Thus the growing season is almost continuous, though summer is the busiest part of the farming year.

10 THE CHINA TYPE CLIMATE

ECONOMIC DEVELOPMENT

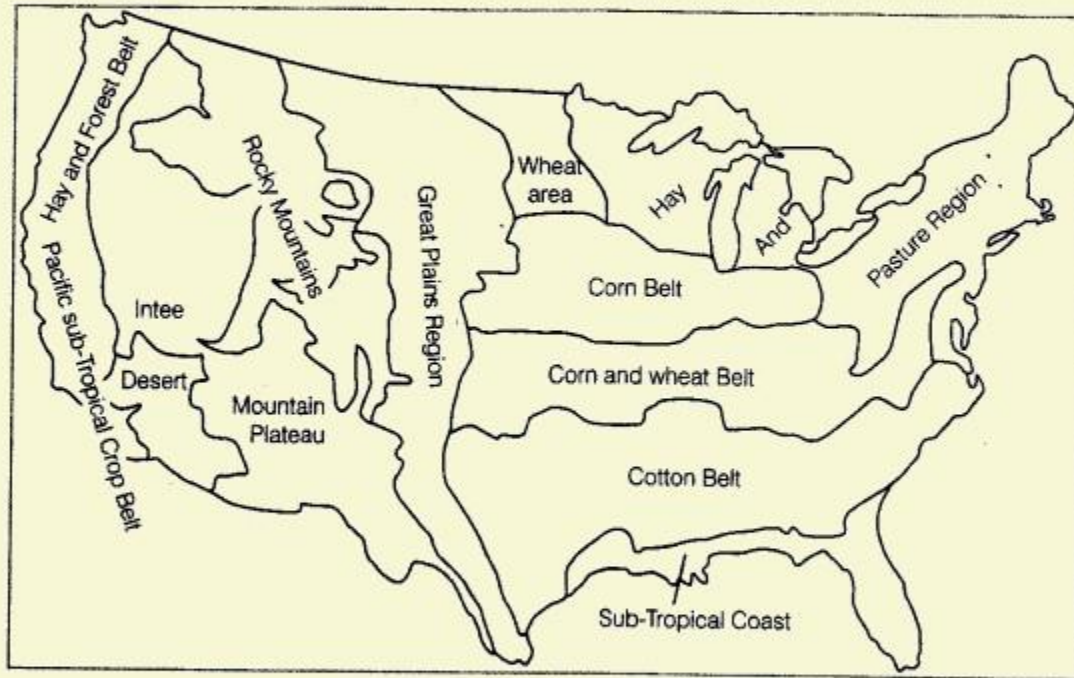


Terraced rice cultivation in China

- 4** Monsoon China together with southern Japan and other parts of the eastern margin climatic zone accounts for almost a third of the world population.
- 5** Food has to be raised to feed the teeming population. The hills are terraced, fields are irrigated, and agriculture is extended to the limits of production.
- 6** It is no exaggeration to say that the temperate monsoon lands are one of the most intensively tilled parts of the earth.

10 THE CHINA TYPE CLIMATE

ECONOMIC DEVELOPMENT



Cotton and Corn belt of USA

7 Besides the widespread cultivation of maize and cotton in the Corn and Cotton Belts of U.S.A, fruit and tobacco are also grown.

8 Rice, tea and mulberries are extensively grown in monsoon China.

9 Elsewhere are found other products of economic importance, e.g. cane sugar in Natal, coffee and maize in South America and dairying in New South Wales and Victoria.

10 Let us now examine some of the regions more closely

10 THE CHINA TYPE CLIMATE

ECONOMIC DEVELOPMENT

FARMING IN MONSOON CHINA

TYPE- 1



There are farmers in China who grow rice and raise fishes in the same fields.

- 1 Undoubtedly this is the **world's greatest rice growing area**.
- 2 A third of the world's rice is grown in China, though the huge population leaves very little for export.
- 3 The Chinese peasants raise '**wet padi**' or '**swamp rice**' in **flooded fields** that call for endless hard labour for the greater part of the year.

10 THE CHINA TYPE CLIMATE

ECONOMIC DEVELOPMENT

FARMING IN MONSOON CHINA

TYPE- 1



Intensive Wet Paddy Cultivation in China

- 4 Farming is usually on a **subsistence basis**.
- 5 Monsoon China has all the ideal conditions for padi cultivation; a warm climate, moderately wet throughout the year, and extensive lowlands with fertile moisture-retentive alluvial soil, which if necessary, can be easily irrigated.
- 6 Besides rice the other important crops are tea, grown for home consumption and mulberry leaves gathered for feeding silk worms, though sericulture is declining.

10 THE CHINA TYPE CLIMATE

ECONOMIC DEVELOPMENT

TYPE- 2

AGRICULTURE IN THE GULF STATES

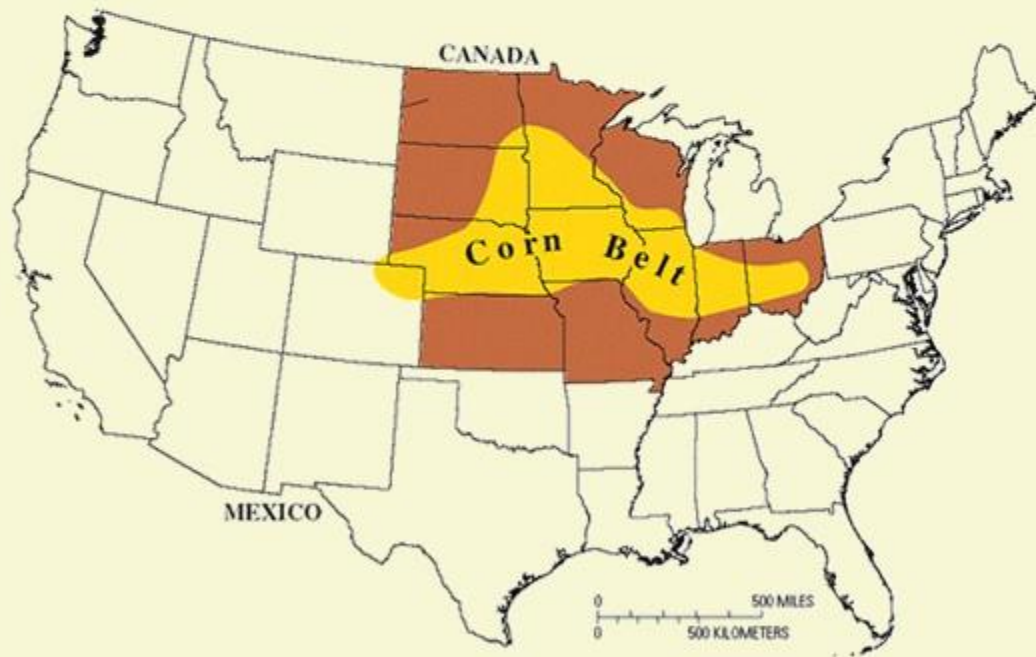
- 1 Agriculture in the Gulf states of America differs from that of monsoon China, though they have a similar climate.
- 2 Lack of population pressure and the urge to export, make rice cultivation a relatively unimportant occupation.
- 3 It is grown only in a few areas in the southern coast-lands of the Mississippi delta.
- 4 American are bread-eaters and one can well imagine how insignificant is rice in the economy of the Gulf states.
- 5 The most important crops are corn, cotton and tobacco.

10 THE CHINA TYPE CLIMATE

ECONOMIC DEVELOPMENT

TYPE- 2

AGRICULTURE IN THE GULF STATES



6 CORN:

- The chief food crop raised is, in fact, corn or maize.
- It is grown right from the Gulf coast to the Mid-west south of the Great lakes, with the greatest concentration in the **Corn Belt of Nebraska, Iowa, Indiana and Ohio.**
- The region accounts for more than half the world's production of corn.
- This is because **most of the corn is used for fattening animals, mostly cattle and pigs.**
- The fattened animals are then sold to the **meat plants in Chicago and Cincinnati to be slaughtered and processed into 'corned beef' or frozen and chilled beef.**

10 THE CHINA TYPE CLIMATE

ECONOMIC DEVELOPMENT

AGRICULTURE IN THE GULF STATES

TYPE- 2

7 COTTON:

- Of the cash crops grown in the Gulf states, none is comparable with cotton.
- In the early days of America, millions of Negroes were brought from Africa as **slave labour for the cotton plantations, because the climate was too hot for the white settlers to harvest the cotton themselves.**
- The Gulf type of climate is undoubtedly the best for cotton growing .
- **Its long, hot growing season with 200 days frost free** and a moderately high temperature of about 75°F permits the crop to grow slowly and mature within six months.
- Besides the problem of soil exhaustion and erosion caused by prolonged cotton cultivation, the most dreaded enemy of the Cotton Belt is the **boll-weevil.**

10 THE CHINA TYPE CLIMATE

ECONOMIC DEVELOPMENT

TYPE- 2

AGRICULTURE IN THE GULF STATES

8 TOBACCO:

- Another interesting crop closely associated with the Gulf type of climate is tobacco, which incidentally is also a native crop of America.
- Though it is cultivated in many parts of the world, and the finished products range from Turkish tobacco to Havana cigars and Malaysian cheroots, there is none so universally known as the **Virginia tobacco**.
- It is the raw material from which most of the world's cigarettes are blended to suit the smoker's taste.
- The **humid atmosphere, the warmth and the well-drained soils of the Gulf states, enable tobacco to be successfully cultivated** in many of the eastern states of U.S.A., e.g. Virginia, Maryland, Georgia, North and South Carolina , Kentucky and Tennessee.
- No less than half the tobacco that enters international trade comes from these states.

10 THE CHINA TYPE CLIMATE

ECONOMIC DEVELOPMENT

AGRICULTURE IN NATAL REGION

TYPE- 3



Cane sugar grown in Natal

- 1 In the coastlands of Natal, cane sugar is the dominant crop, followed by cotton and tobacco in the interior.
- 2 Maize is extensively cultivated.
- 3 But in comparison with the maize yield of the Corn Belt of U.S.A., the African yield is rather low, often only half.

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